Omer Bobrowski

Professor in Mathematical Data Science, QMUL · AI Researcher, Meta FAIR

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ACADEMIC POSITIONS

Professor in Mathematical Data Science 2024–present
School of Mathematical Sciences, Queen Mary University of London

Visiting Professor 2025–present
Faculty of Electrical and Computer Engineering, Technion – Israel Institute of Technology

Associate Professor 2021–2025
Faculty of Electrical and Computer Engineering, Technion – Israel Institute of Technology

Senior Lecturer 2022–2024
School of Mathematical Sciences, Queen Mary University of London

Assistant Professor 2016–2021
Faculty of Electrical and Computer Engineering, Technion – Israel Institute of Technology

Visiting Assistant Professor 2015–2016
Department of Mathematics, Duke University

Elliott Assistant Research Professor 2012–2015
Department of Mathematics, Duke University

EDUCATION

PhD in Electrical Engineering Technion – Israel Institute of Technology 2008–2012
Thesis: *Algebraic topology of random fields and complexes* · Advisor: Prof. Robert J. Adler

MSc in Electrical Engineering, *summa cum laude* Technion 2004–2008
Thesis: *Real time spike train decoding by neural networks* · Advisors: Prof. Ron Meir, Prof. Yonina Eldar

BA in Mathematics and Computer Science, *summa cum laude* 2001–2004
The Open University, Israel

Visiting Student 2009
Department of Mathematics, University of Chicago

PROFESSIONAL EXPERIENCE

Meta FAIR (Fundamental AI Research) — Research Scientist, part-time 2026–present
Research on AI agents for scientific research.

Israeli Defense Forces — External Consultant 2004–2005
Software engineering and algorithms.

Israeli Defense Forces, Intelligence Corps — Military Service 2001–2004
R&D in software engineering and algorithms; analysis, development and implementation of complex software systems across a variety of platforms and needs.

RESEARCH GRANTS

Mathematical Foundations of Intelligence: An "Erlangen Programme" for AI (hub) 2024–2029
EPSRC · co-Investigator, theme lead · £10M

Topological Phase Transitions in Stochastic Geometry 2024–2027
EPSRC · Principal Investigator · £400k

Universality in Topological Data Analysis 2024–2027
Leverhulme Trust, Research Project Grant · Principal Investigator · £330k

Statistical Challenges in Applied Topology 2019–2023
Israel Science Foundation · Principal Investigator · $225k

SCHOLARSHIPS AND AWARDS

Fellow of the Higher Education Academy (FHEA)	2024
QMUL Faculty of Science and Engineering Excellence in Research Prize	2023
A Distinction for Excellence in Teaching, Technion	2021
Muriel and David Jacknow Award for Excellence in Teaching, Technion	2012
Sandor Szego Awards for Excellence in Teaching, Technion	2010, 2011
Adams Fellowship for Outstanding Doctoral Students, The Israel Academy of Sciences and Humanities	2010–2012
Vivian Konigsberg Award for Excellence in Teaching, Technion	2008, 2011, 2012
Miriam and Aaron Gutwirth Memorial Fellowship, Technion	2007
Dean's List (*cum laude*), The Open University	2003
President's List (*magna cum laude*), The Open University	2001, 2002, 2004

EDITORIAL DUTIES

Associate Editor 2025–present
Journal of Applied and Computational Topology & Geometry (ACTG)

Associate Editor 2025–present
Journal of Applied and Computational Topology (JACT)

Guest Editor, Special Issue on Random Topology 2022–2024
Journal of Applied and Computational Topology (JACT)

Editorial board of reviewers 2020–present
Journal of Machine Learning Research (JMLR)

Referee for journals and conferences

Referee for numerous journals and conferences, including Proceedings of the National Academy of Sciences (PNAS), Journal of the European Mathematical Society (JEMS), Advances in Mathematics, Probability Theory and Related Fields, Annals of Applied Probability, Annals of Statistics, Transactions of the AMS, Israel Journal of Mathematics, IEEE Symposium on Foundations of Computer Science (FOCS), ACM-SIAM Symposium on Discrete Algorithms (SODA), Journal of Machine Learning Research (JMLR) and IEEE Transactions on Information Theory.

Referee for funding agencies

Austrian Science Fund (FWF) · Israel Science Foundation (ISF) · German Research Foundation (DFG) · Hungarian Academy of Sciences (MTA) · National Science Centre Poland · The Royal Society · US–Israel Binational Science Foundation (BSF)

PUBLICATIONS

37. **A Universal Law of Large Numbers for Extreme Cycles in Random Čech Complexes** 2026
O. Bobrowski and P. Skraba — submitted. arXiv:2609.19474

36. **Universal Topological Statistics on Triangulated Singular Spaces** 2026
U. Lim, O. Bobrowski, P. Skraba — submitted. arXiv:2607.27535

35. **A Universal Nearest-Neighbor Estimator for Intrinsic Dimensionality** 2026
E. J. Ong, O. Bobrowski, G. Reinert, P. Skraba — to appear in IEEE Transactions on Pattern Analysis & Machine Intelligence. arXiv:2603.10493

34. **Sharp Phase Transition for the Formation of Infinite Tubes** 2026
S. Kanazawa, O. Bobrowski, and P. Skraba — submitted. arXiv:2605.14910

33. **Central Limit Theorems for Local Functionals of Dynamic Point Processes** 2026
E. Onaran, O. Bobrowski, and R. Adler — The Electronic Journal of Probability 31, 1–31

32. **Universality in Random Persistent Homology and Scale-Invariant Functionals** 2026
O. Bobrowski and P. Skraba — The Communications of the AMS 6, 378–423

31. **Sharp Phase Transitions for k-Fold Coverage Using Morse Theory** 2025
Y. Reani and O. Bobrowski — submitted. arXiv:2510.20329

30. **Morse Theory for the k-NN Distance Function (journal version)** 2025
Y. Reani and O. Bobrowski — Discrete & Computational Geometry 75(2), 491–510

29. **Morse Theory for the k-NN Distance Function (conference version)** 2024
Y. Reani and O. Bobrowski — 40th International Symposium on Computational Geometry (SoCG)

28. **Cluster-Persistence for Weighted Graphs** 2023
O. Bobrowski and P. Skraba — Entropy 25(12), Special Issue on Models, Topology and Inference of Multilayer and Higher-Order Networks

27. **Functional CLTs for Local Statistics of Spatial Birth-Death Processes in the Thermodynamic Regime** 2023
E. Onaran, O. Bobrowski, and R. Adler — The Annals of Applied Probability 33(5), 3958–3986

26. **A Coupled Alpha Complex** 2023
Y. Reani and O. Bobrowski — Journal of Computational Geometry 14(1), 221–256

25. **A Universal Null-Distribution for Topological Data Analysis** 2023
O. Bobrowski and P. Skraba — Scientific Reports 13(1), 12274

24. **Cycle Registration in Persistent Homology with Applications in Topological Bootstrap** 2023
Y. Reani and O. Bobrowski — IEEE Transactions on Pattern Analysis & Machine Intelligence 45(5), 5579–5593

23. **Poisson Process Approximation Under Stabilization and Palm Coupling** 2022
O. Bobrowski, M. Schulte, and D. Yogeshwaran — Annales Henri Lebesgue 5, 1489–1534

22. **Homological Connectivity in Random Čech Complexes** 2022
O. Bobrowski — Probability Theory and Related Fields 183(3–4), 715–788

21. **Unsupervised Geometric and Topological Approaches for Cross-Lingual Sentence Representation** 2022
S. H. Meirom, O. Bobrowski — ACL Proceedings of the 7th Workshop on Representation Learning for NLP, 173–183

20. **Alarm Sound Detection Using Topological Signal Processing** 2022
T. Fireaizen, S. Ron, O. Bobrowski — IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 211–215

19. **Random Simplicial Complexes: Models and Phenomena (book chapter)** 2022
O. Bobrowski and D. Krioukov — Understanding Complex Systems: High-Order Systems, 59–96, Springer

18. **On the Spectrum of Dense Random Geometric Graphs** 2022
K. Adhikari, R. Adler, O. Bobrowski, and R. Rosenthal — The Annals of Applied Probability 32(3), 1734–1773

17. **Homological Percolation: The Formation of Giant k-Cycles** 2022
O. Bobrowski and P. Skraba — International Mathematics Research Notices 2022(8), 6186–6213

16. **Joint Geometric and Topological Analysis of Hierarchical Datasets** 2021
L. Aloni, O. Bobrowski, and R. Talmon — Machine Learning and Knowledge Discovery in Databases, ECML PKDD, 478–493

15. **Probabilistic Convergence and Stability of Random Mapper Graphs** 2021
A. Brown, O. Bobrowski, E. Munch, and B. Wang — Journal of Applied and Computational Topology 5, 99–140

14. **Homological Percolation and the Euler Characteristic** 2020
O. Bobrowski and P. Skraba — Physical Review E 101, 032404

13. **Convergence of Persistence Diagrams for Topological Crackle** 2020
T. Owada and O. Bobrowski — Bernoulli 26(3), 2275–2310

12. **Random Čech Complexes on Riemannian Manifolds** 2019
O. Bobrowski and G. Oliveira — Random Structures & Algorithms 54(3), 373–412

11. **Topology of Random Geometric Complexes: A Survey** 2018
O. Bobrowski and M. Kahle — Journal of Applied and Computational Topology 1(3–4), 331–364

10. **Maximally Persistent Cycles in Random Geometric Complexes** 2017
O. Bobrowski, M. Kahle, and P. Skraba — The Annals of Applied Probability 27(4), 2032–2060

9. **On the Vanishing of Homology in Random Čech Complexes** 2017
O. Bobrowski and S. Weinberger — Random Structures & Algorithms 51(1), 14–51

8. **Topological Consistency via Kernel Estimation** 2017
O. Bobrowski, S. Mukherjee, and J. Taylor — Bernoulli 23(1), 288–328

7. **The Topology of Probability Distributions on Manifolds** 2015
O. Bobrowski and S. Mukherjee — Probability Theory and Related Fields 161(3–4), 651–686

6. **Crackle: The Homology of Noise** 2014
R. J. Adler, O. Bobrowski, and S. Weinberger — Discrete & Computational Geometry 52(4), 680–704

5. **Distance Functions, Critical Points and the Topology of Random Čech Complexes** 2014
O. Bobrowski and R. J. Adler — Homology, Homotopy and Applications 16(2), 311–345

4. **Euler Integration of Gaussian Random Fields and Persistent Homology** 2012
O. Bobrowski and M. S. Borman — Journal of Topology and Analysis 4(01), 49–70

3. **Persistent Homology for Random Fields and Complexes** 2010
R. J. Adler, O. Bobrowski, M. S. Borman, E. Subag, and S. Weinberger — Borrowing Strength: Theory Powering Applications, IMS Collections Vol. 6, 124–143

2. **Bayesian Filtering in Spiking Neural Networks: Noise, Adaptation and Multisensory Integration** 2009
O. Bobrowski, R. Meir, and Y. C. Eldar — Neural Computation 21(5), 1277–1320

1. **A Neural Network Implementing Optimal State Estimation Based on Dynamic Spike Train Decoding** 2007
O. Bobrowski, R. Meir, S. Shoham, and Y. C. Eldar — Advances in Neural Information Processing Systems (NIPS) 20, 145–152

TEACHING EXPERIENCE

Queen Mary University of London, School of Mathematical Sciences 2021–present

Postgraduate	Advanced Machine Learning; Computational Statistics with R
LTCC	Introduction to Random Topology
Coursera	Module development: Storing, Manipulating and Visualising Data

Tel-Aviv University, School of Mathematical Sciences 2018–2019

Postgraduate	Topological Methods in Statistics (national graduate statistics programme)

Technion – Israel Institute of Technology, Electrical and Computer Engineering 2017–2021

Undergraduate	Random Signals
Postgraduate	Special Topics in Probability and Stochastic Processes; Foundations of Stochastic Processes; Topological Methods in Engineering

Duke University, Mathematics and Statistics 2012–2016

Undergraduate	Multivariate Calculus; Probability and Statistics in Engineering
Postgraduate	Topics in Probability

Technion – Israel Institute of Technology, Mathematics and ECE (teaching assistant, during PhD) 2008–2012

Tutorials	Introduction to Probability; Complex Functions; Fourier Analysis; Digital Signal Processing

ACADEMIC SUPERVISION

Postdoctoral Scholars

Dr. Qiquan Wang 2025–present

Dr. Shu Kanazawa 2025–present

Dr. Eng-Jon Ong 2025–present

Dr. Uzu Lim 2024–present

Dr. Parker Duncan 2024–2026

Dr. Efe Onaran — now Postdoctoral Researcher, University of Pennsylvania, USA 2020–2023

Dr. Khartic Adhikari — now Assistant Professor, Mathematics, IISER Bhopal, India 2018–2020

Dr. Sunder Ram Krishnan — now Assistant Professor, Computing, Amrita Vishwa Vidyapeetham, India 2017–2018

PhD Students

Ziv Maayan (co-supervised with Prof. Ronen Talmon, Technion) 2026–present

Flavio Gualtieri 2025–present

Matilde Muzzolini 2025–present

Yohai Reani 2021–2026

MSc Students (two-year research thesis)

Shaked Meirom 2020–2023

Lior Aloni Abuhasira 2018–2021

Yohai Reani 2018–2020

CONFERENCE AND SEMINAR ORGANISATION

Scientific board — Erlangen AI Hub Conference, University of Oxford, UK 2026

Organiser — Applied Geometry and Topology (AGT) Network 2025–present

Organiser — London–Oxford Topological Data Analysis (TDA) Seminar 2024–present

Organiser — London–Oxford–Paris TDA Workshop, City University, London, UK 2025

Scientific board — Erlangen AI Hub Conference, Queen Mary University of London, UK 2025

Organiser — Joint Israeli Probability Seminar (JIPS), weekly online seminar 2020–2022

Organiser — Technion Probability Seminar, weekly, Technion, Israel 2018–2020

Organiser — Haifa Probability School 3 (HPS3), Technion, Israel 2020

Invited session organiser — Stochastic Processes and their Applications (SPA), Chicago, IL 2019

Contributed session organiser — Stochastic Processes and their Applications (SPA), Gothenburg, Sweden 2018

Scientific committee — Algebraic Topology – Methods, Computation and Science (ATMCS7), Turin, Italy 2016

TUTORIALS AND WORKSHOPS

Project Supervision: Cycle Matching for High-Dimensional Neural Activation Patterns 2025
London Geometry and Machine Learning (LOGML), Imperial College London, UK

Invited Mini-Course: Random Topology – Theory and Applications 2025
TDA in stochastic geometry and image processing, Aarhus Institute of Advanced Studies, Denmark

Invited Mini-Course: Random Simplicial Complexes 2021
High Dimensional Spatial Random Systems Workshop, Hausdorff Research Institute for Mathematics (HIM), Bonn, Germany (online)

Invited Mini-Course: Random Simplicial Complexes 2015
Research School on Computational Algebraic Topology, Clay Mathematics Institute & London Mathematical Society, Oxford University, UK

Invited Mini-Course: Stochastic Topology 2015
TDA-School (Topological Data Analysis), CIMAT, Guanajuato, Mexico

INVITED TALKS

48. Algebraic Topology: Methods, Computation & Science (ATMCS), Stockholm, Sweden PLENARY 2027
47. Highlights in Probability, Hebrew University, Israel 2026
46. Order and Disorder in Random Geometries: Hyperuniformity and TDA, Castle of Reisensburg, Germany 2026
45. Curves and Surfaces, mini-symposium on TDA, Saint-Malo, France 2026
44. Point Processes and Their Applications, Lille, France 2026
43. Geometric Methods in Probability, QMUL, London, UK 2026
42. The Topology of Human and Synthetic Minds, King's College London, UK 2025
41. Workshop on Random Simplicial Complexes and Hypergraphs, Nottingham Trent University, UK 2025
40. The Geometric Realization of AATRN (Applied Algebraic Topology Research Network), IMSI, Chicago, IL 2025
39. Dynamics, Topology and Computations, Banach Center, Będlewo, Poland PLENARY 2025

38. Stochastic Geometry in Action, University of Bath, Bath, UK 2024
37. Theory and Applications of Hypergraphs, Isaac Newton Institute, London, UK 2024
36. Quantitative Topology and Related Areas, Simons Foundation, New York, NY 2024
35. Computational Persistence Workshop, Purdue University, West Lafayette, IN 2023
34. New Perspectives in Extreme Values, IUC, Dubrovnik, Croatia 2023
33. Randomness in Topology and Its Applications, IMSI, Chicago, IL 2023
32. TDA Week, 10th International Congress on Industrial and Applied Mathematics (ICIAM), Kyoto, Japan KEYNOTE 2023
31. Spring School on Geometric Methods in Data Science, ORT Braude, Carmiel, Israel 2023
30. LMS Workshop on Applied Topology and Robot Motion Planning, QMUL, London, UK 2023
29. Geometry, Topology and Statistics in Data Sciences, Institut Henri Poincaré, Paris, France 2022
28. Adaptive and High-Dimensional Spatio-Temporal Methods for Forecasting, CIRM, Marseille, France 2022
27. IMS Annual Meeting, London, UK 2022
26. TDA Week Workshop, CREST, Japan Science and Technology Agency, Japan 2022
25. Thematic Einstein Semester “Geometric and Topological Structure of Materials”, TU Berlin, Germany 2021
24. VII Mexican Workshop on Geometric and Topological Data Analysis, CIMAT, Guanajuato, Mexico 2021
23. Annual Meeting of the Centre for TDA, Liverpool, UK 2021
22. SIAM Conference on Applied Algebraic Geometry, Bern, Switzerland 2019
21. Geometric Data Analysis, Stevanovich Center — University of Chicago, Chicago, IL 2019
20. International Symposium on Computational Geometry (SoCG), Budapest, Hungary 2018
19. Bridging Statistics and Sheaves, IMA, University of Minnesota, Minneapolis, MN 2018
18. Topological Data Analysis Meets Symplectic Topology, Tel Aviv University, Israel 2018
17. Topological Data Analysis and Related Topics, AIMR, Tohoku University, Sendai, Japan 2017
16. Probabilistic Methods in Topology, CRM, Université de Montréal 2016
15. Stochastic Topology and Thermodynamic Limits, ICERM, Brown University, Providence, RI 2016
14. Computational and Methodological Statistics (CMStatistics), London, UK 2015
13. Stochastic Processes and Random Fields: Geometry and Fine Properties, Technion, Haifa, Israel 2015
12. Extreme Value Analysis (EVA), Ann Arbor, MI 2015
11. Geometry and Data Analysis, Stevanovich Center — University of Chicago, Chicago, IL 2015
10. Southeast Geometry Seminar XXV, University of Tennessee, Knoxville, TN 2014
9. Joint Statistical Meetings (JSM), Boston, MA 2014
8. Algebraic Topology – Methods, Computation and Science (ATMCS6), Vancouver, Canada 2014
7. TDA Workshop, Statistical and Applied Mathematical Sciences Institute (SAMSI), NC 2014
6. IMA Workshop on Topological Data Analysis, University of Minnesota, Minneapolis, MN 2013
5. Joint Statistical Meetings (JSM), Montréal, Canada 2013
4. Applied Topology Conference, Będlewo, Poland 2013
3. Annual Meeting of the Canadian Applied & Industrial Mathematics Society (CAIMS), Quebec, Canada 2013
2. The Israel Statistical Association Annual Conference, Netanya, Israel 2012
1. AMS Special Session on Applied Topology, University of Akron, Akron, OH 2012

OTHER CONFERENCES AND TALKS (last five years)

13.	Probability Seminar, Technion, Israel	2025
12.	Special Guest Seminar, Université Côte d'Azur, Nice, France	2024
11.	Probability Seminar, Queen Mary University of London, UK	2024
10.	Bernoulli-IMS 11th World Congress in Probability and Statistics, Bochum, Germany	2024
9.	Internal School Colloquium, Queen Mary University of London, UK	2024
8.	Mathematics Research Seminar, Brunel University of London, UK	2023
7.	Probability Seminar, University of Bath, Bath, UK	2023
6.	Yahoo Research, US (online)	2023
5.	London Analysis and Probability Seminar, UCL, London, UK	2022
4.	Applied Geometry and Topology Seminar, CIMAT, Guanajuato, Mexico	2022
3.	Algebraic Topology: Methods, Computation and Science (ATMCS10), Oxford, UK	2022
2.	Industrial and Applied Mathematics Seminar, Oxford University, Oxford, UK	2022
1.	Applied Mathematics Seminar, Haifa University, Haifa, Israel	2021